

Agentic AI for Developers

Understanding Agentic AI



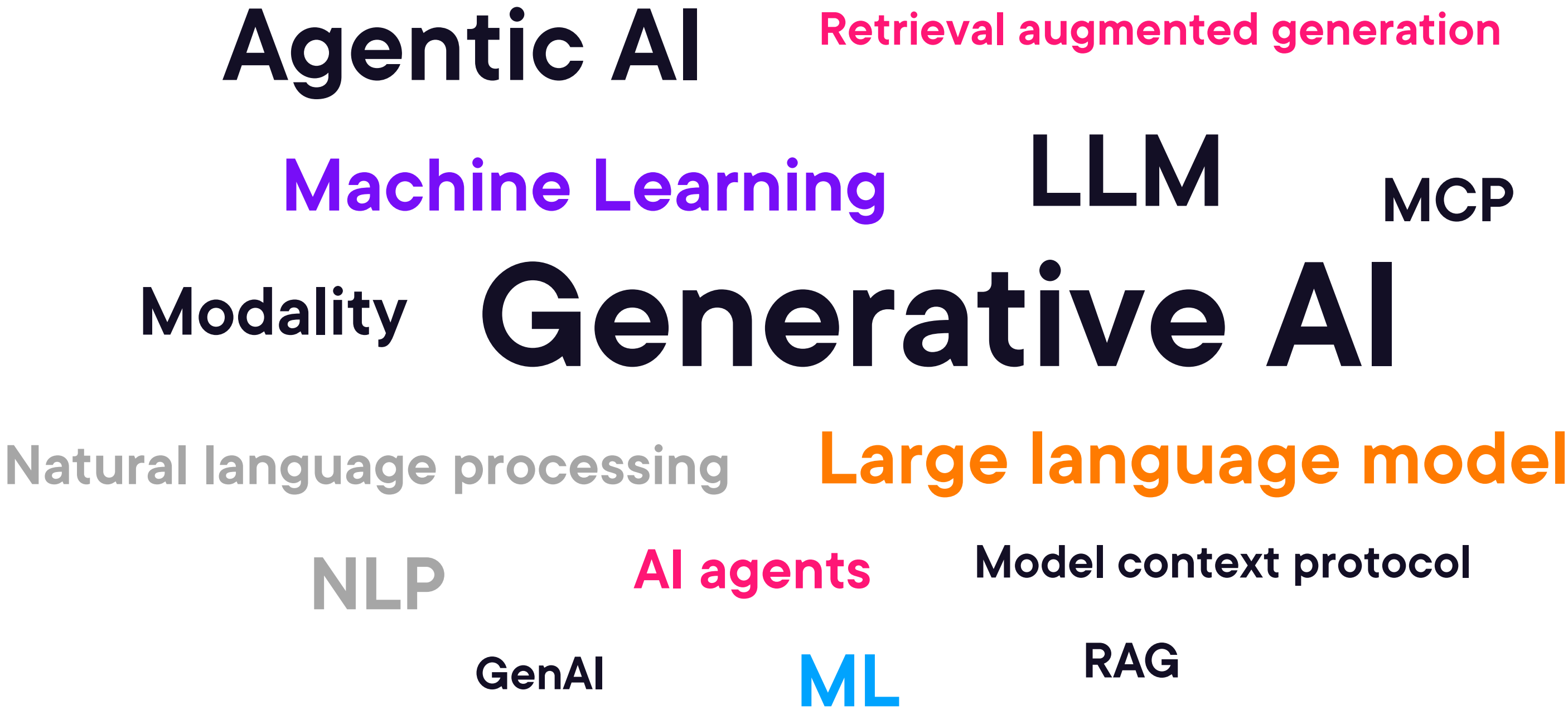
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Engineering Lead

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AI Buzzwords



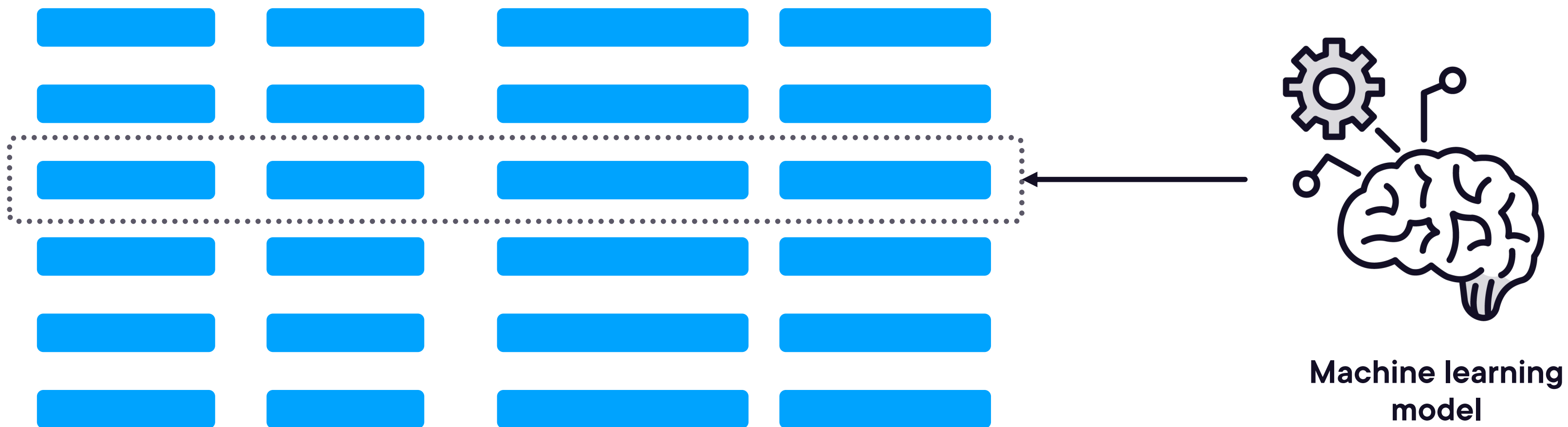
M L



Machine

Learning







Natural Language

Unstructured data

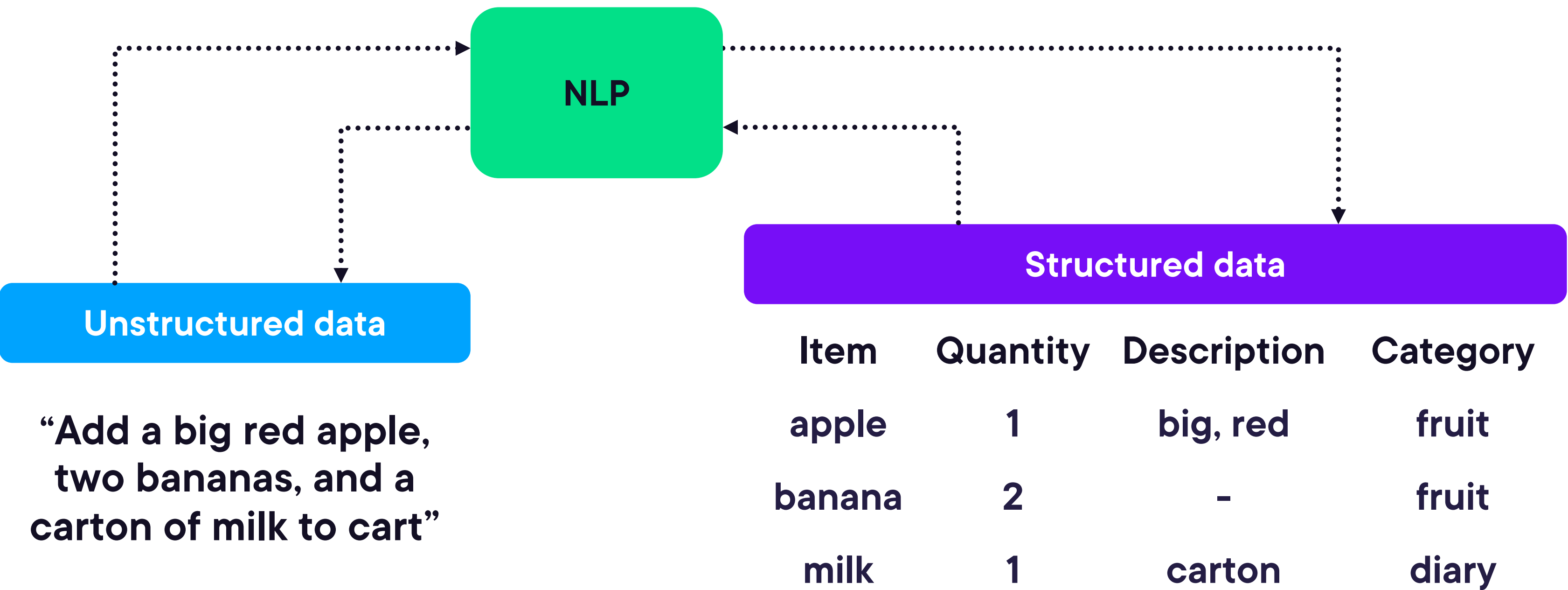
**“Add a big red apple,
two bananas, and a
carton of milk to cart”**

Structured data

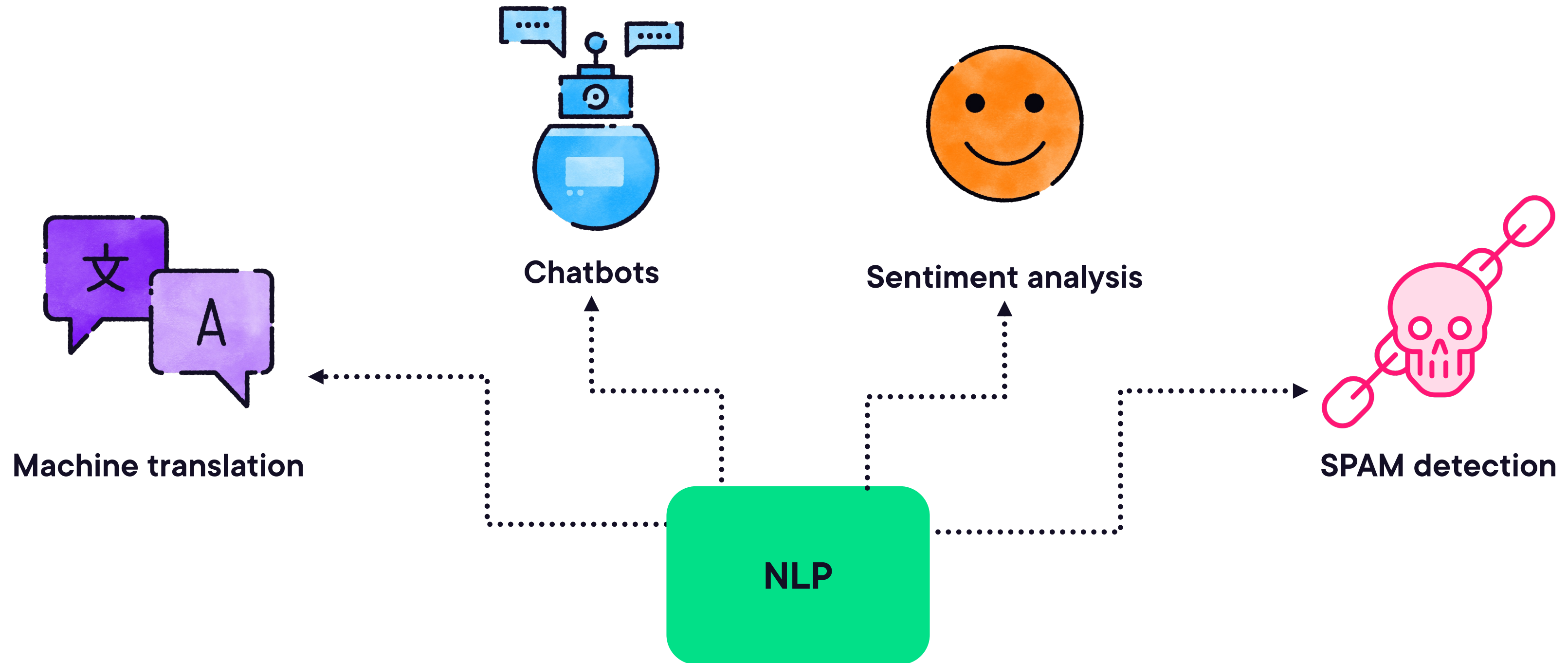
Item	Quantity	Description	Category
apple	1	big, red	fruit
banana	2	-	fruit
milk	1	carton	diary



Natural Language



Natural Language Processing

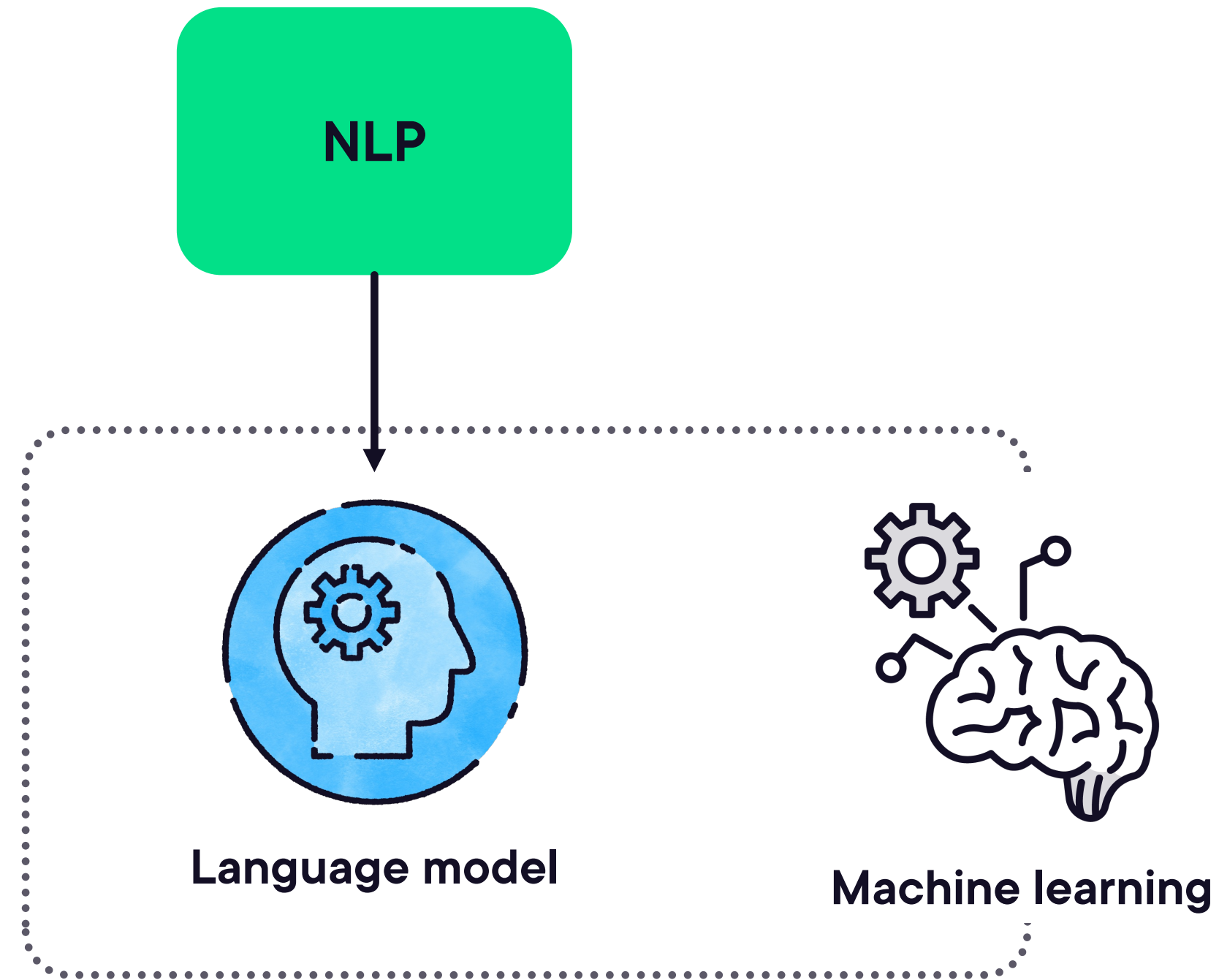




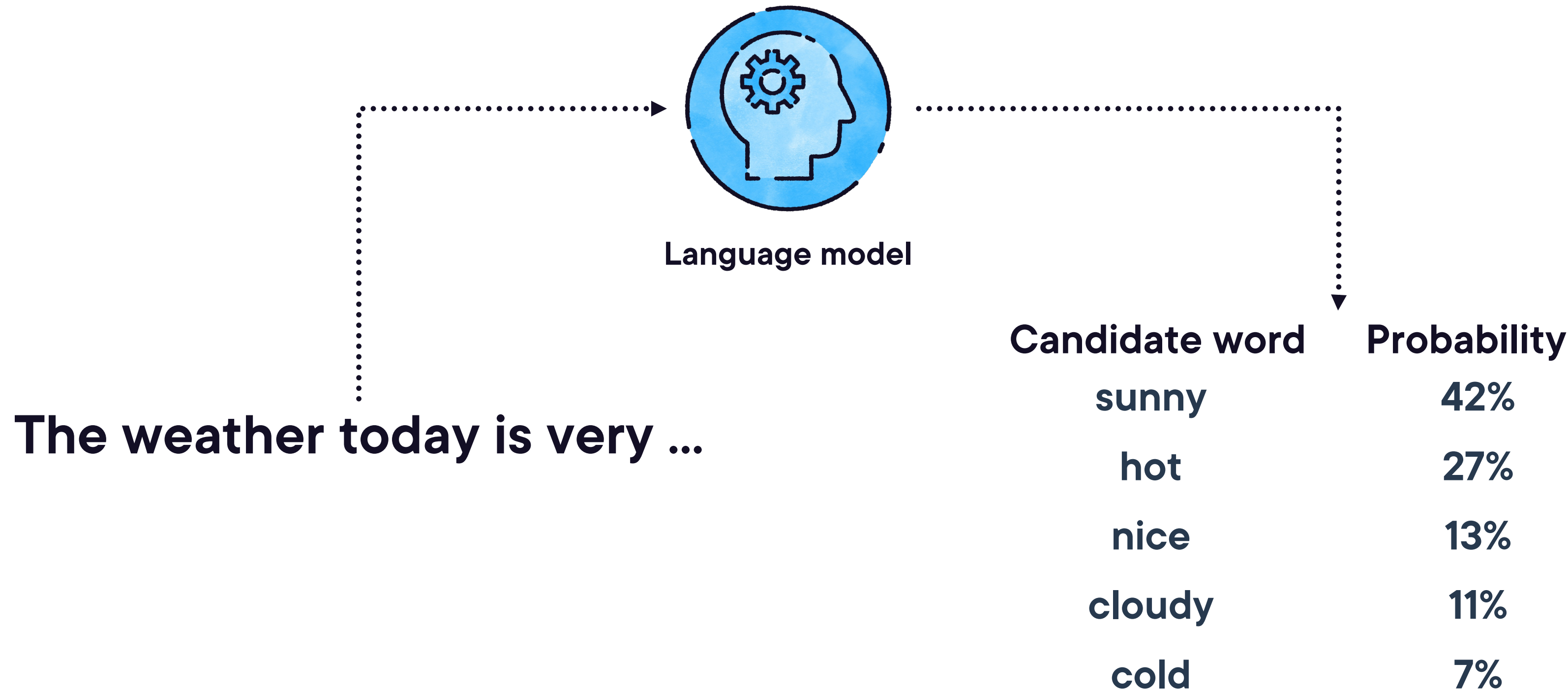
Large Language Models



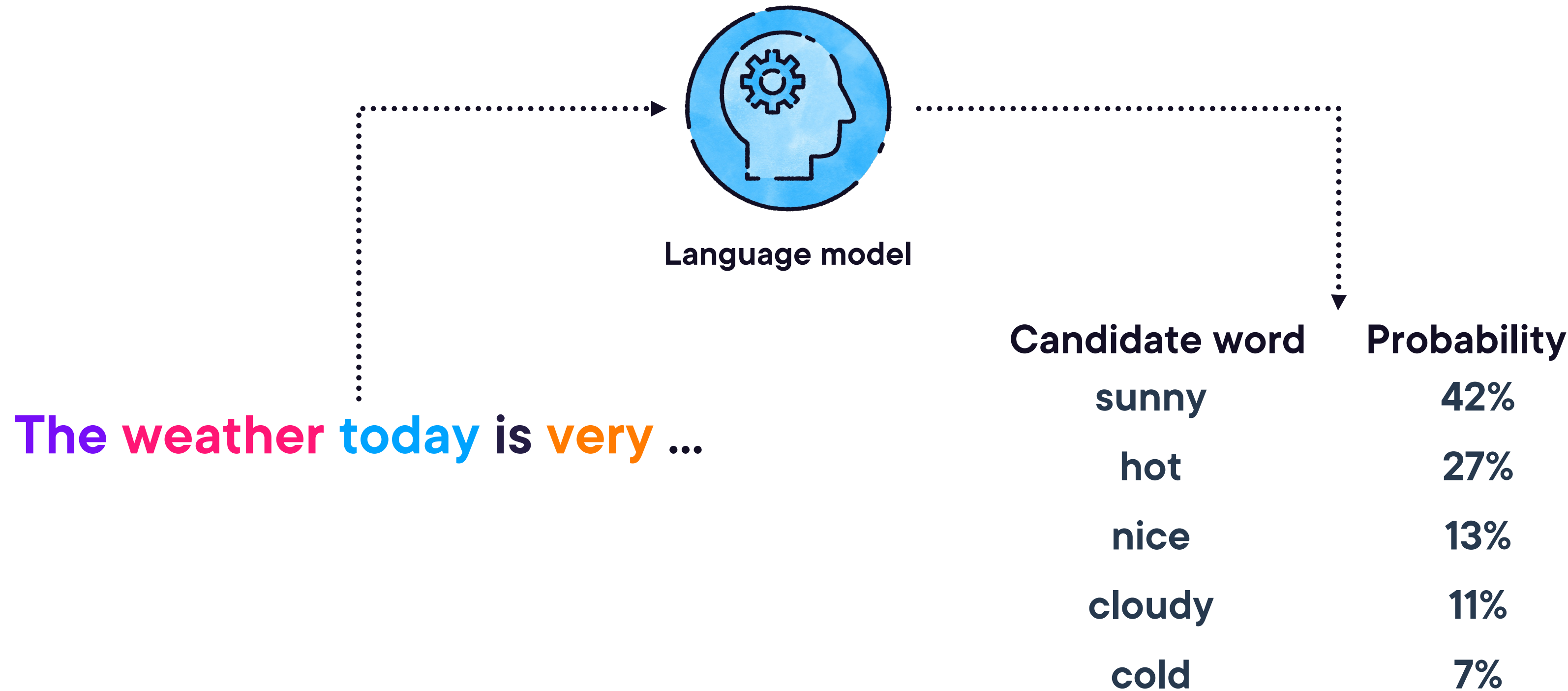
Natural Language Processing



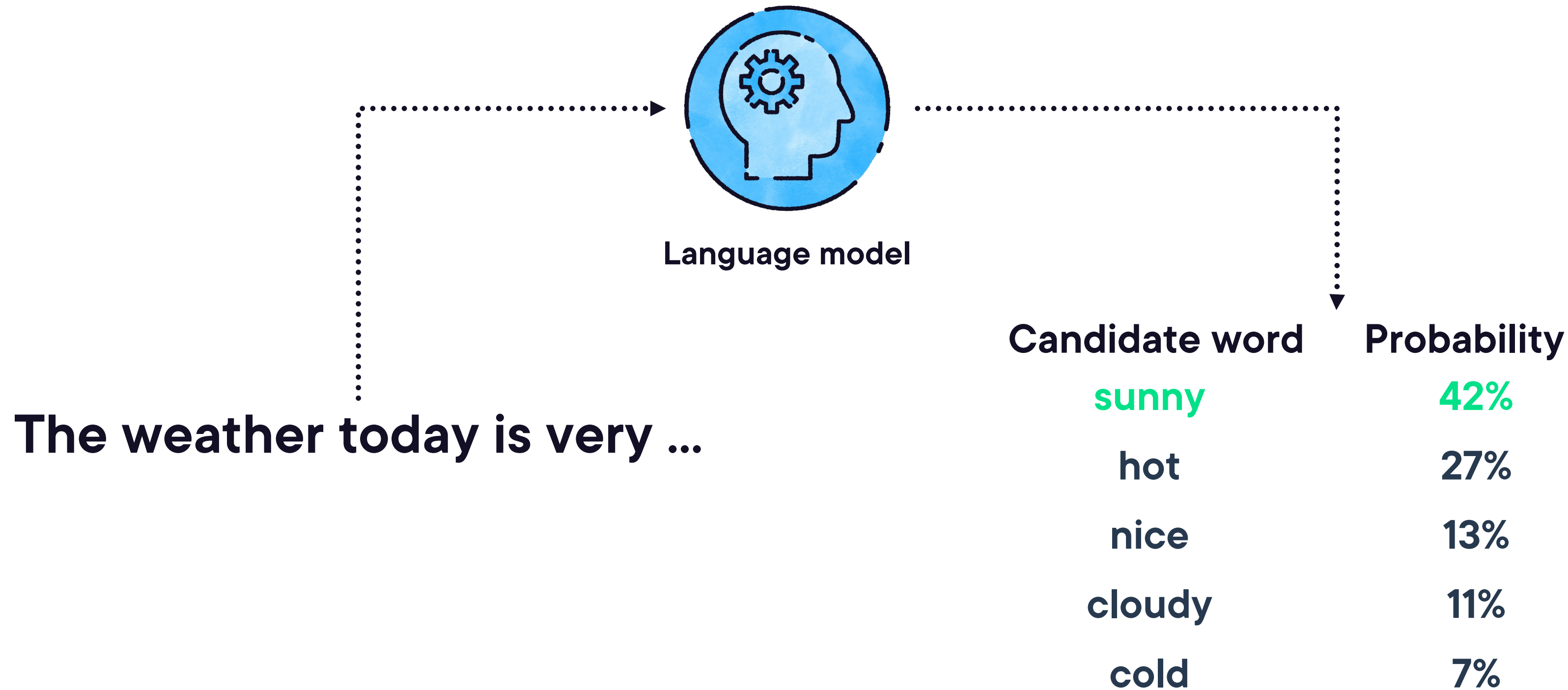
Language Model



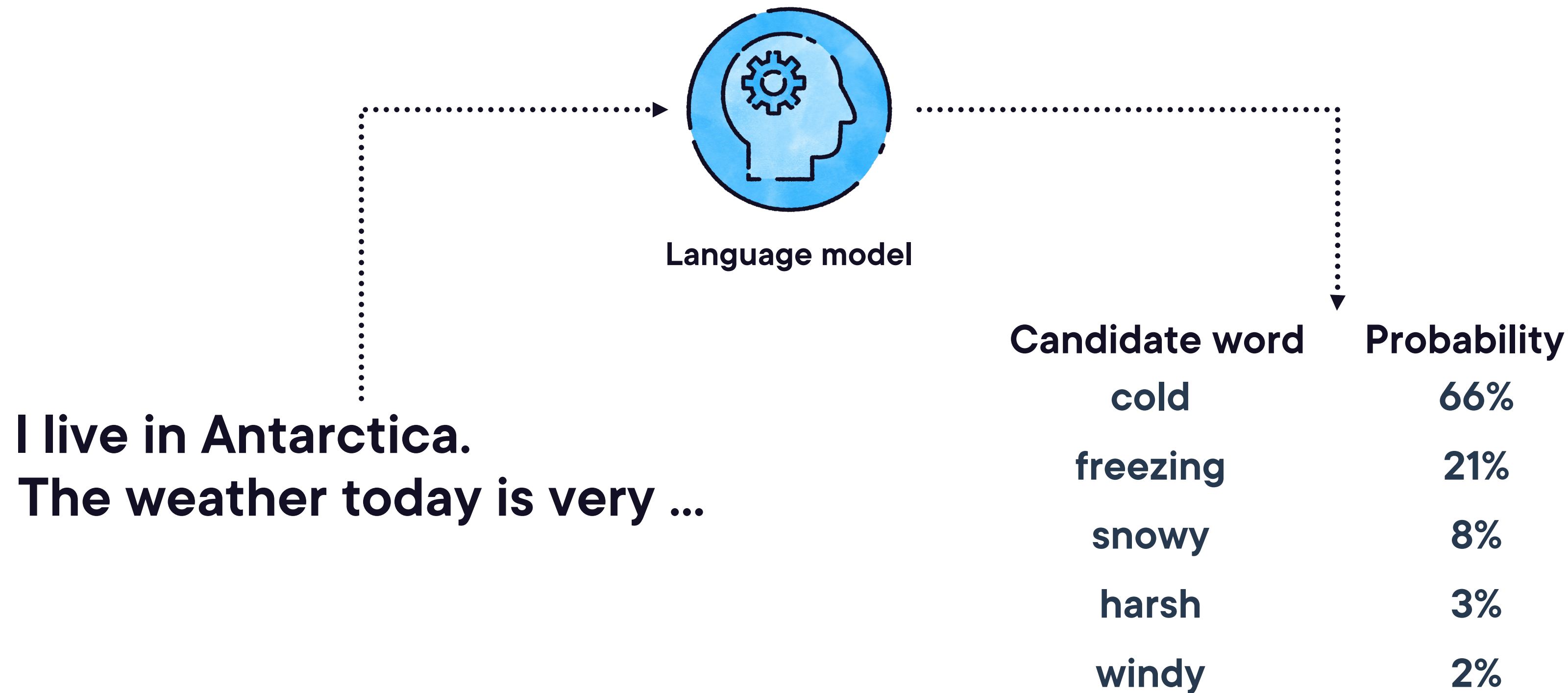
Language Model



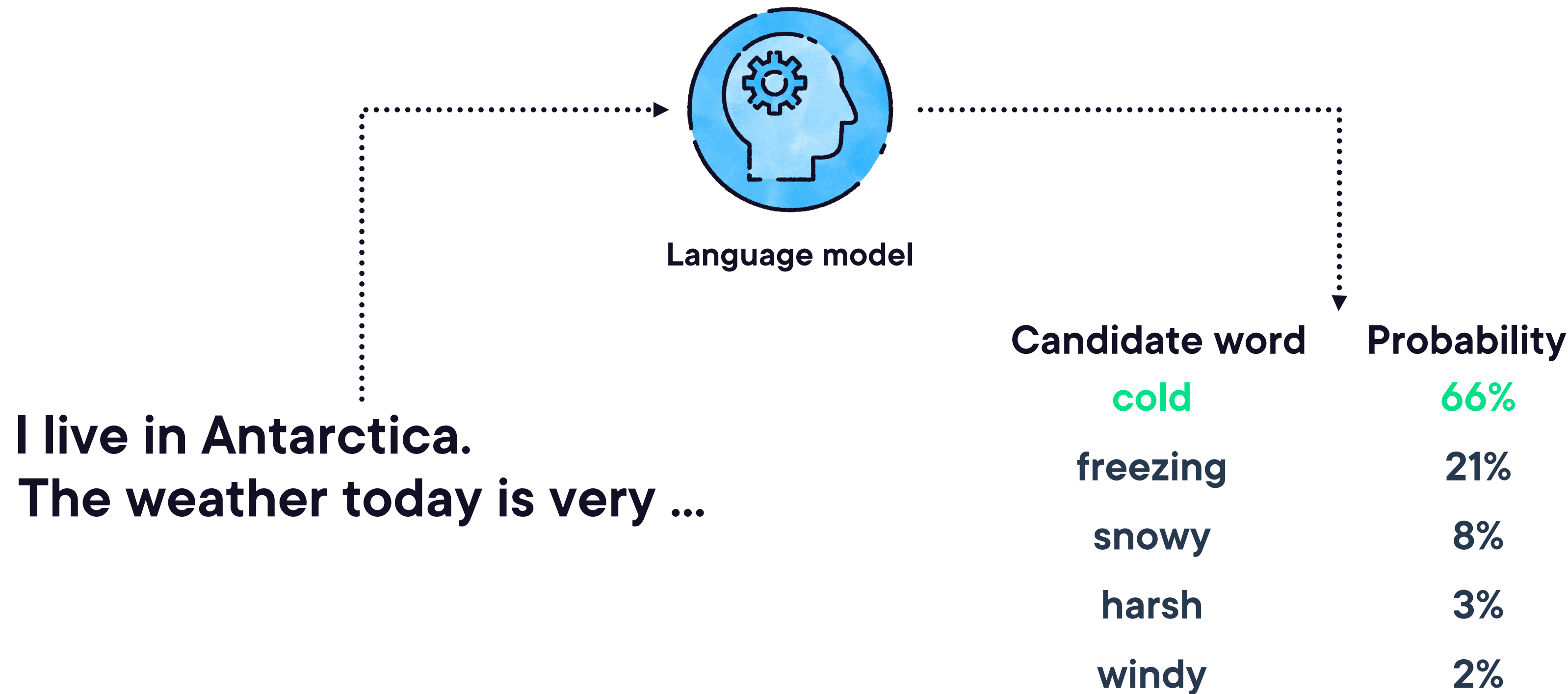
Language Model



Language Model



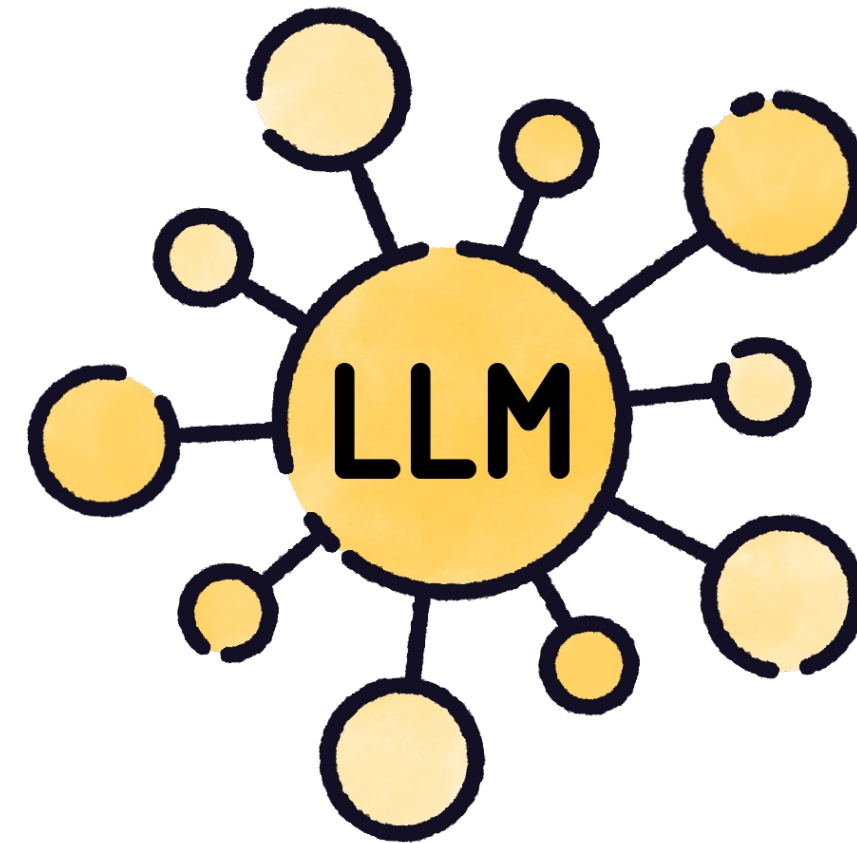
Language Model



Language Model



Language
model



Large language
model

Large Language Models

Language model

Modest datasets

Various architectures

Trainable on CPU or single GPUs

Basic prediction or classification

Task specific

vs.

Large language model

Internet-scale data

Primarily transformer-based

Compute clusters

Complex reasoning

General purpose



Code

```
function calculateTotalPrice(items, taxRate) {  
  let total = 0;  
  
  items.forEach(item => {  
    total += item.price;  
  });  
}
```



Code

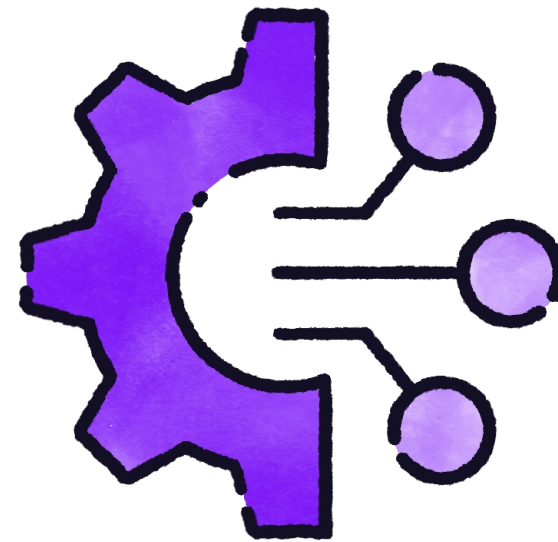
```
function calculateTotalPrice(items, taxRate) {  
  let total = 0;  
  
  items.forEach(item => {  
    total += item.price;  
  });  
  
  total = total + total * taxRate;  
  
  return total;  
}
```



Ways AI Can Help Us



**Increase engineering
productivity**



**Improve operational
efficiency**



Innovate products

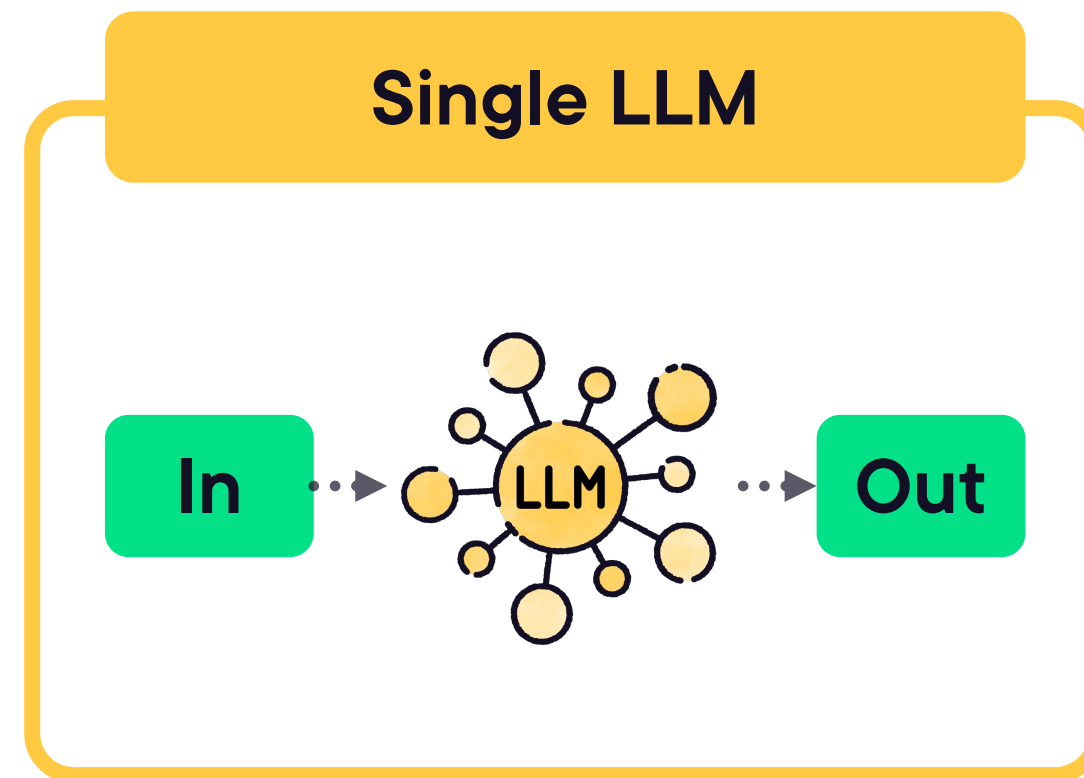




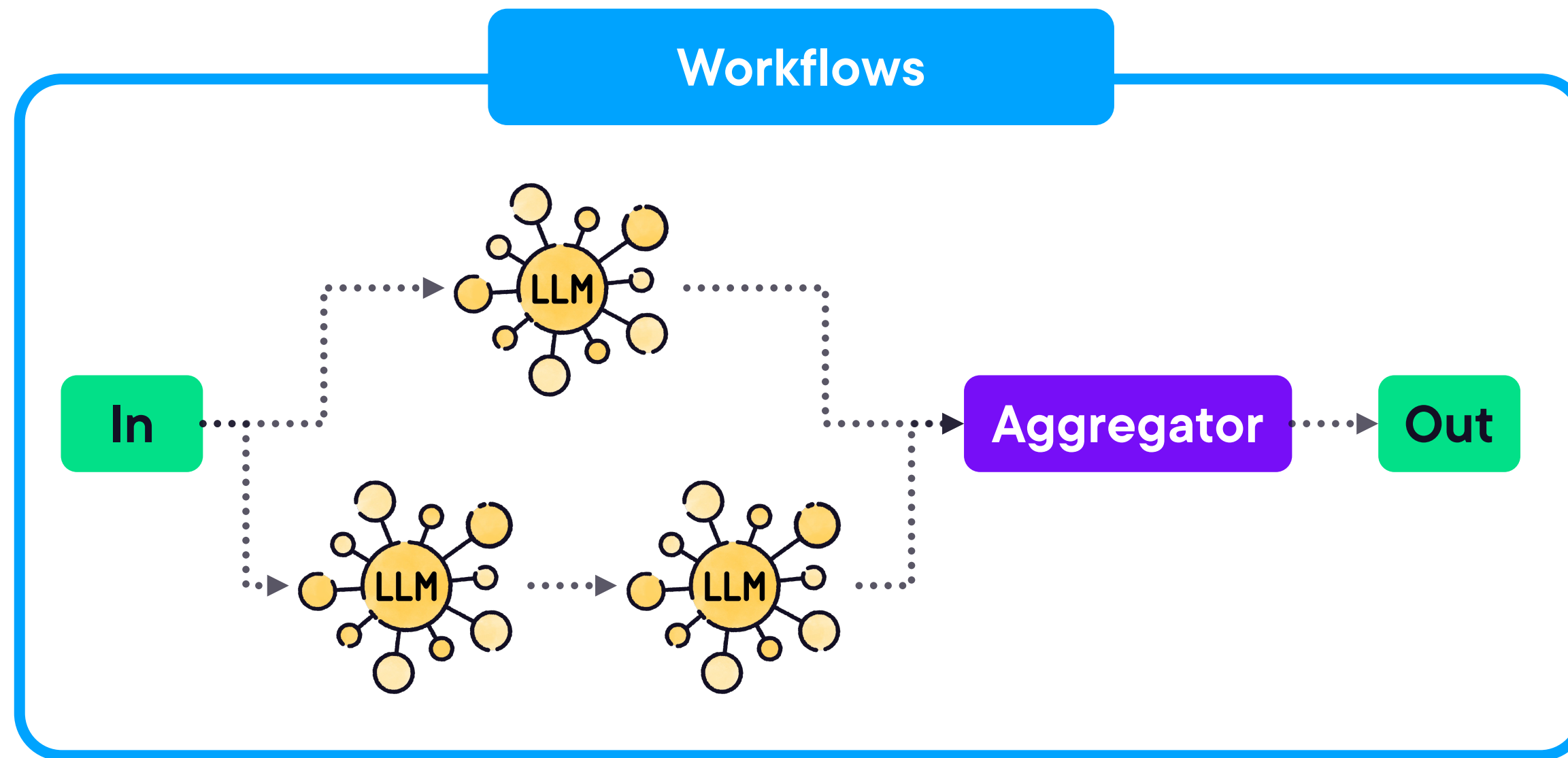
Agentic AI



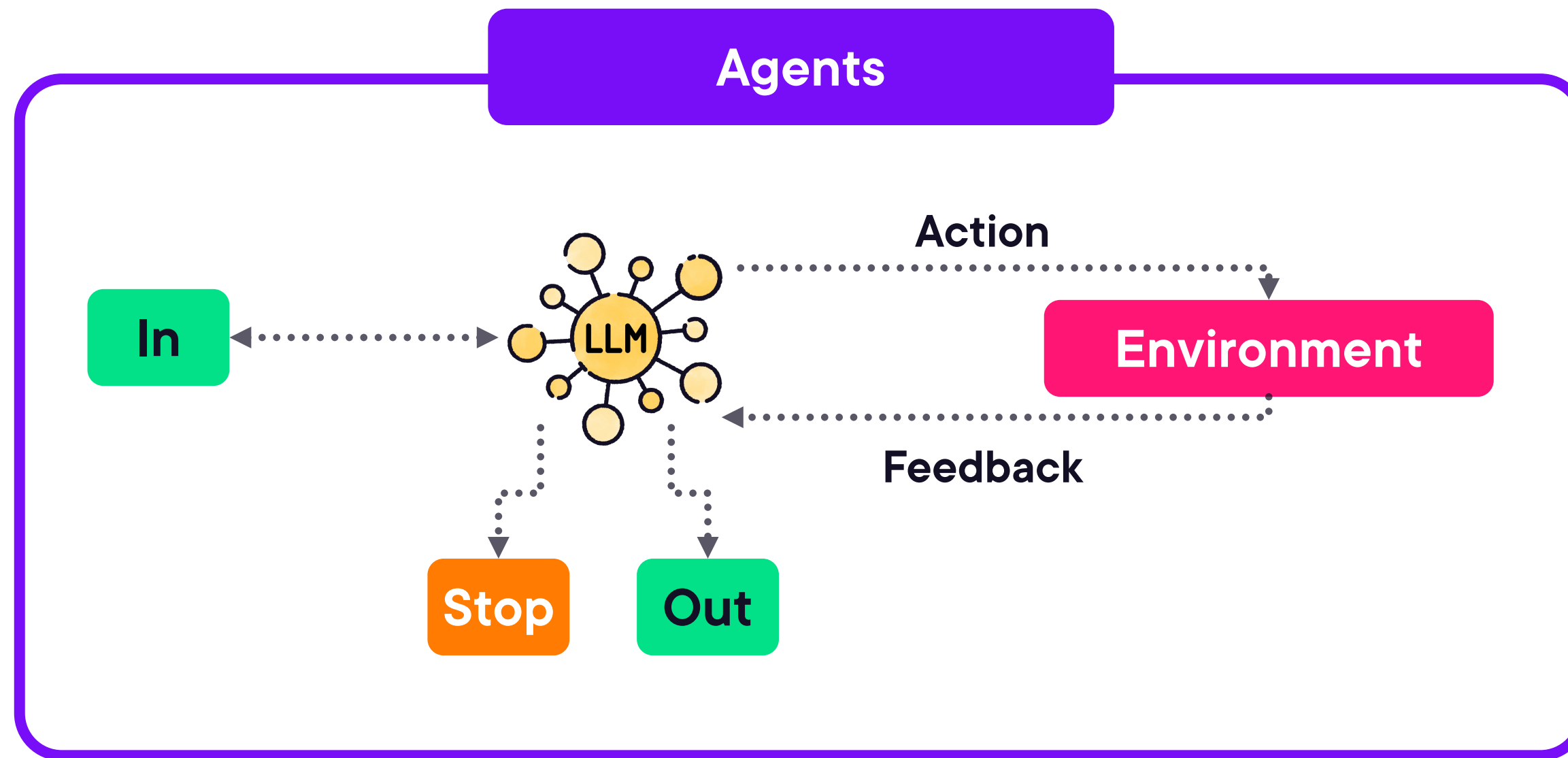
Agentic AI Evolution



Agentic AI Evolution



Agentic AI Evolution

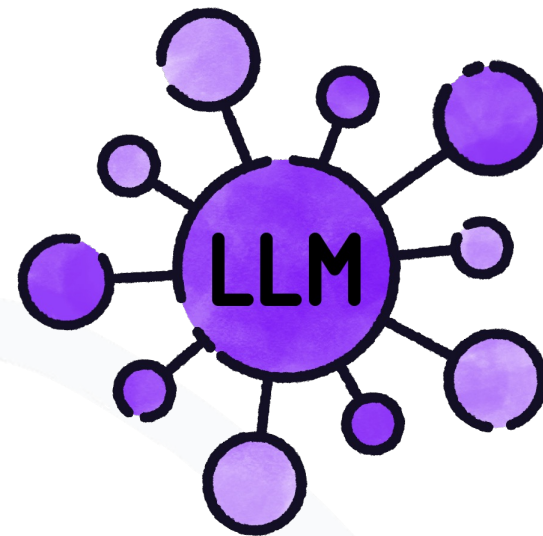


Agentic AI

Perceive



Reason



Act



Learn



Agentic AI

Perceive

Reason

Act

Learn

Plan

Act

Adapt



Agentic AI System

Autonomous execution

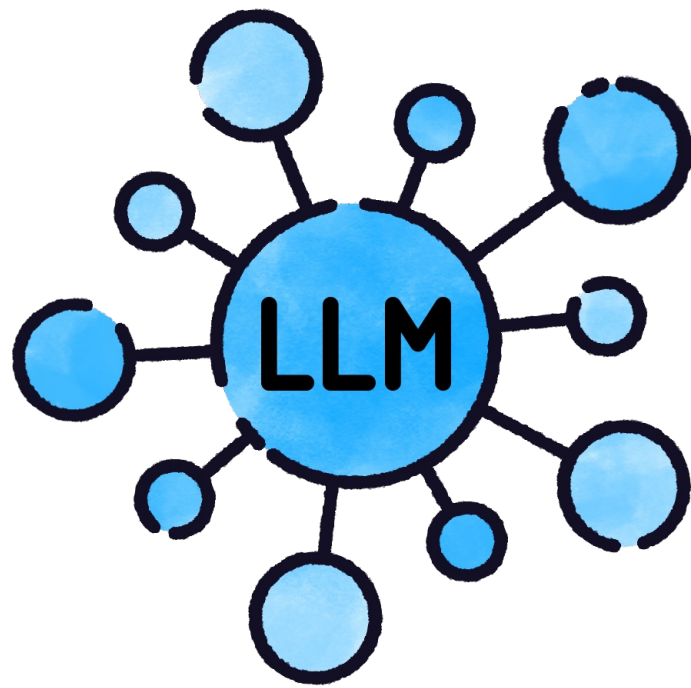
Proactivity

Goal-oriented

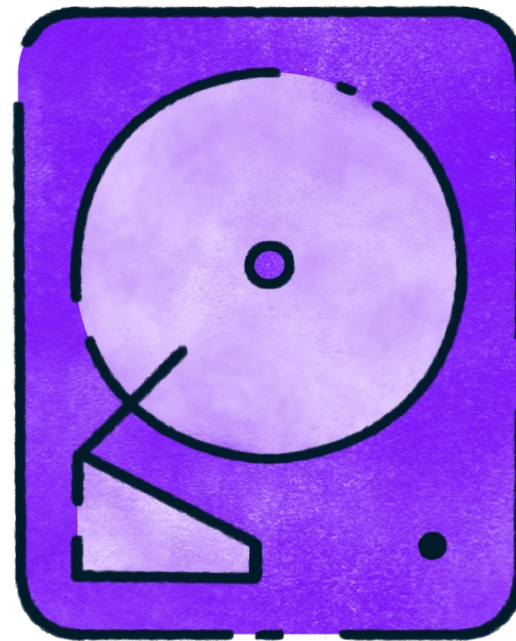
Collaboration



AI Agents



Reasoning engine



Memory



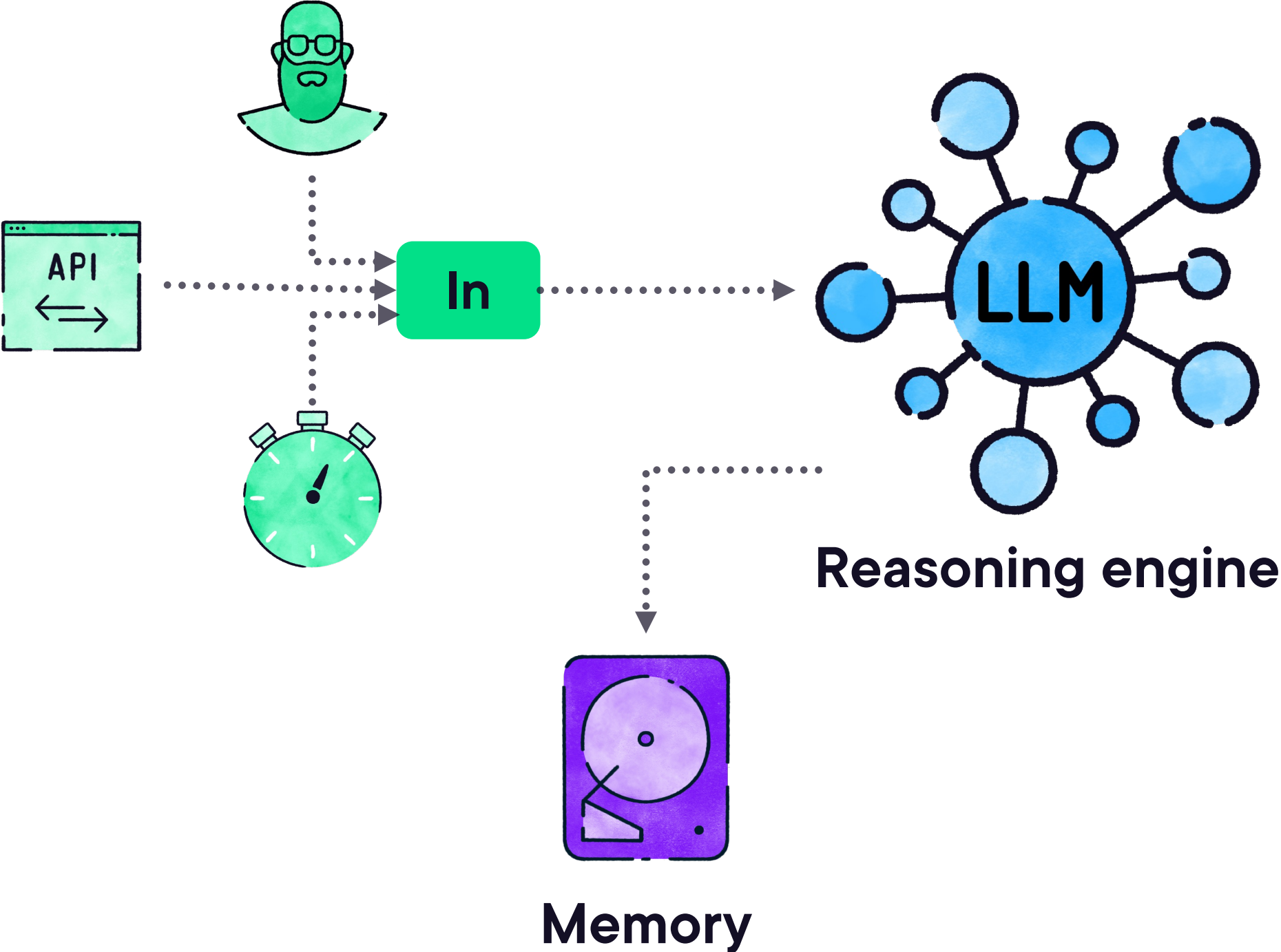
Tools



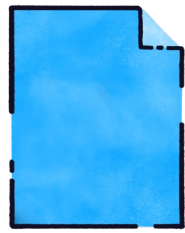
Memory and Tools



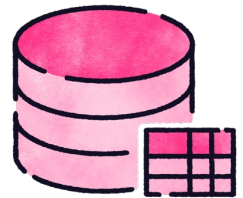
AI Agents



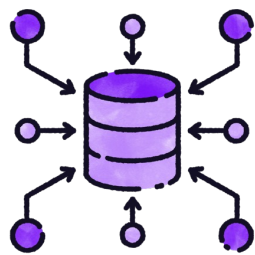
Memory Types



Files



Relational databases



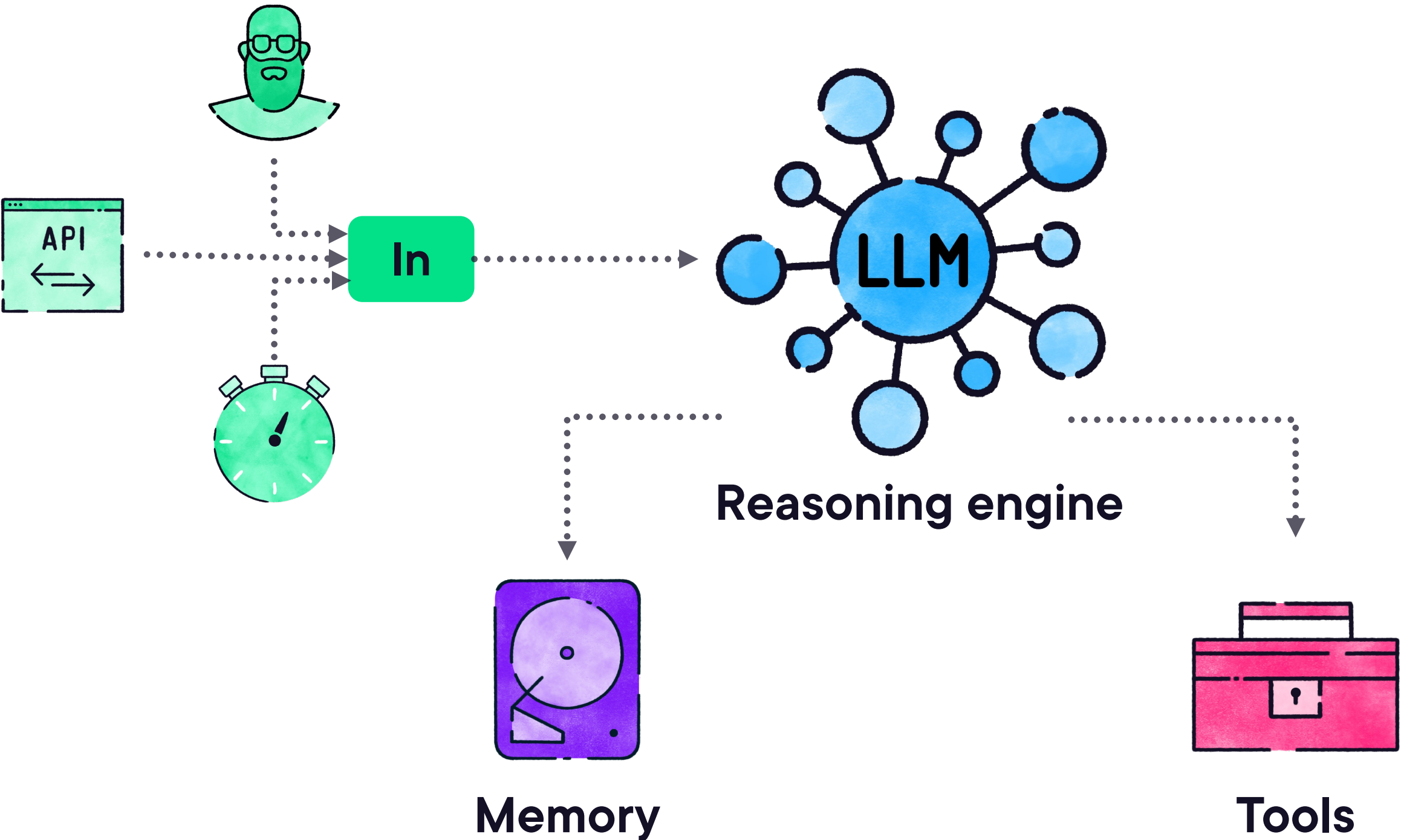
NoSQL datastores



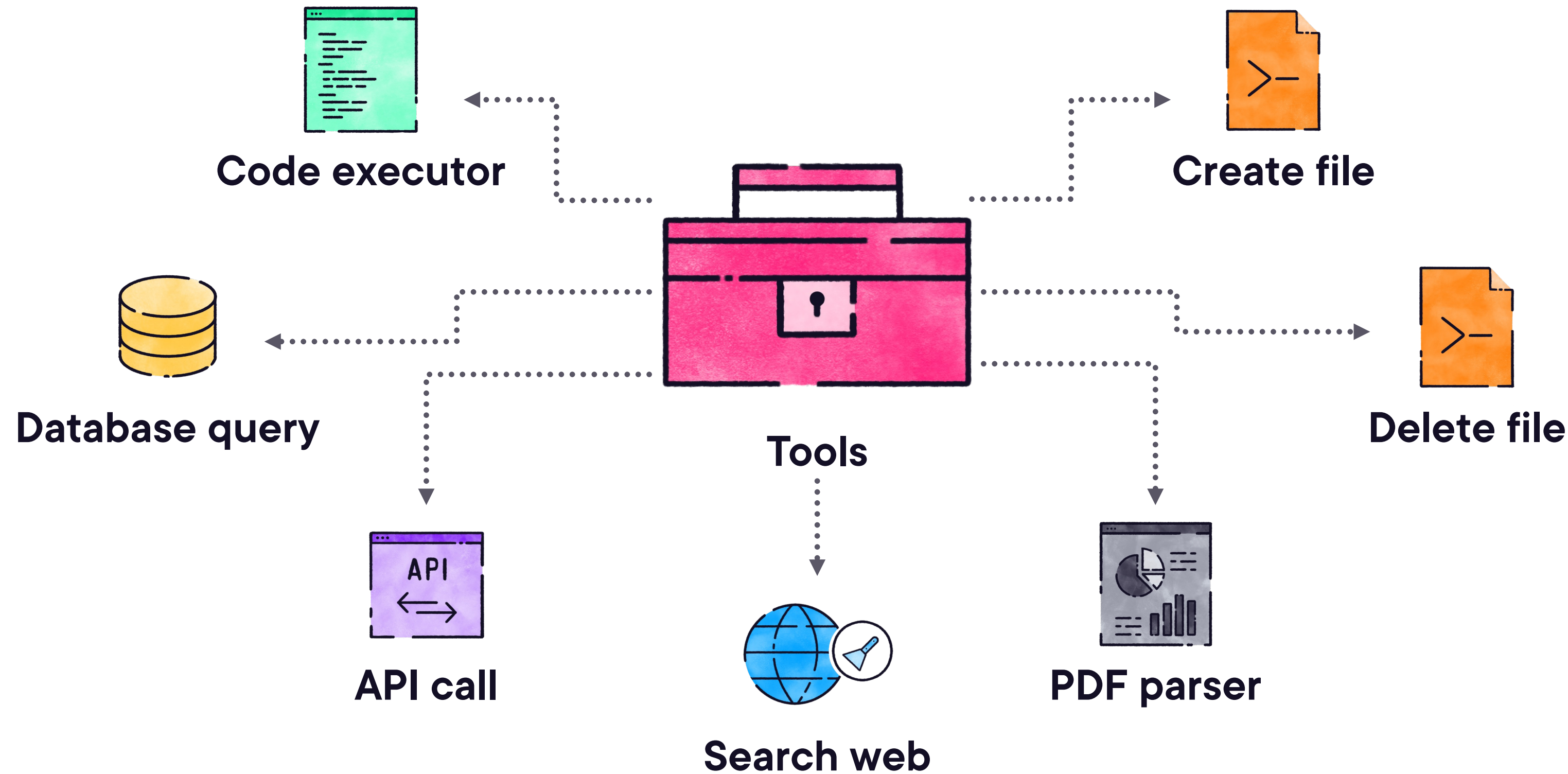
Dedicated AI memory solutions



AI Agents



Tools

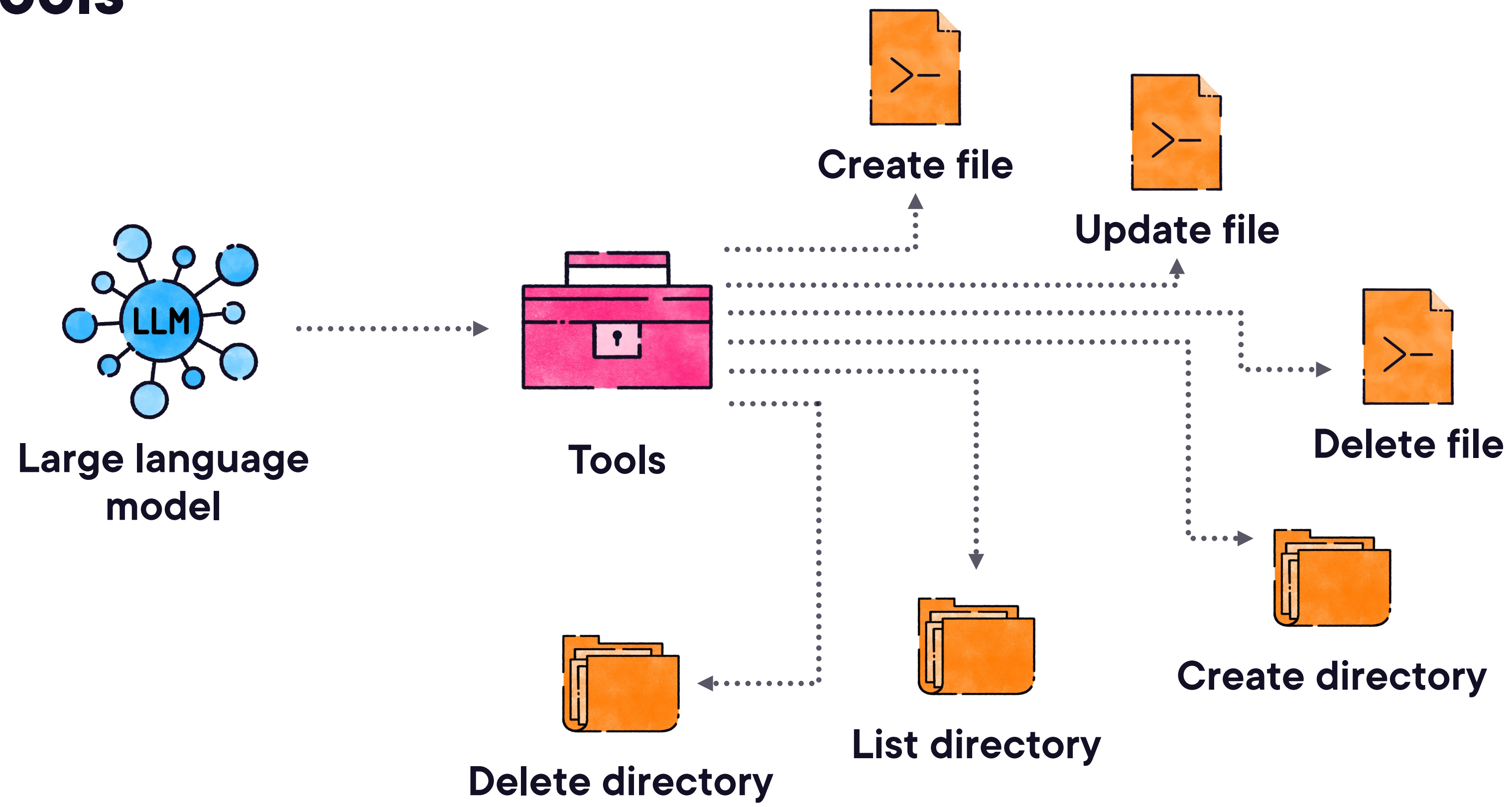




MCP



Tools

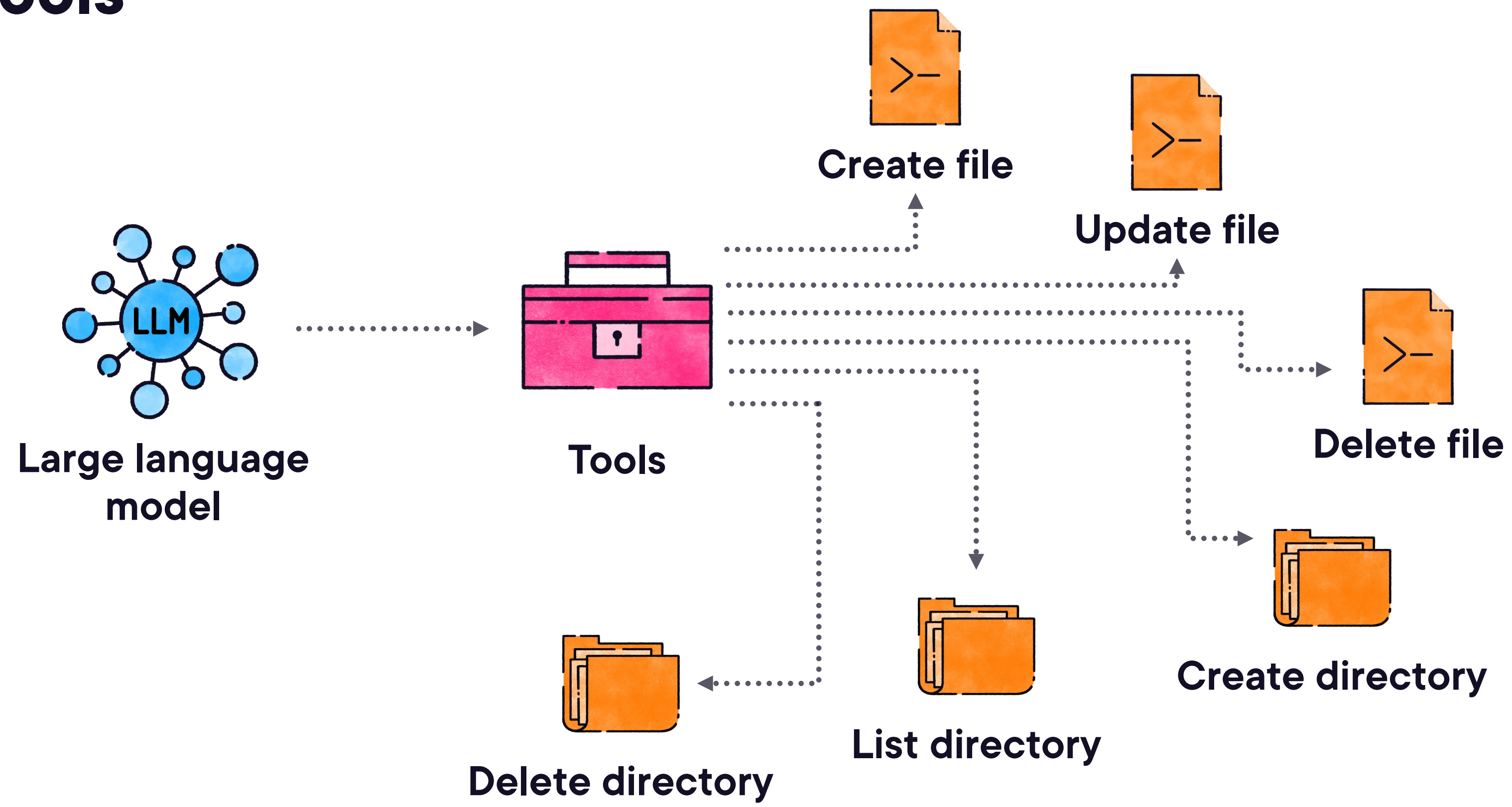


Model Context Protocol (MCP)

MCP is an open protocol that standardizes how applications provide context to LLMs.



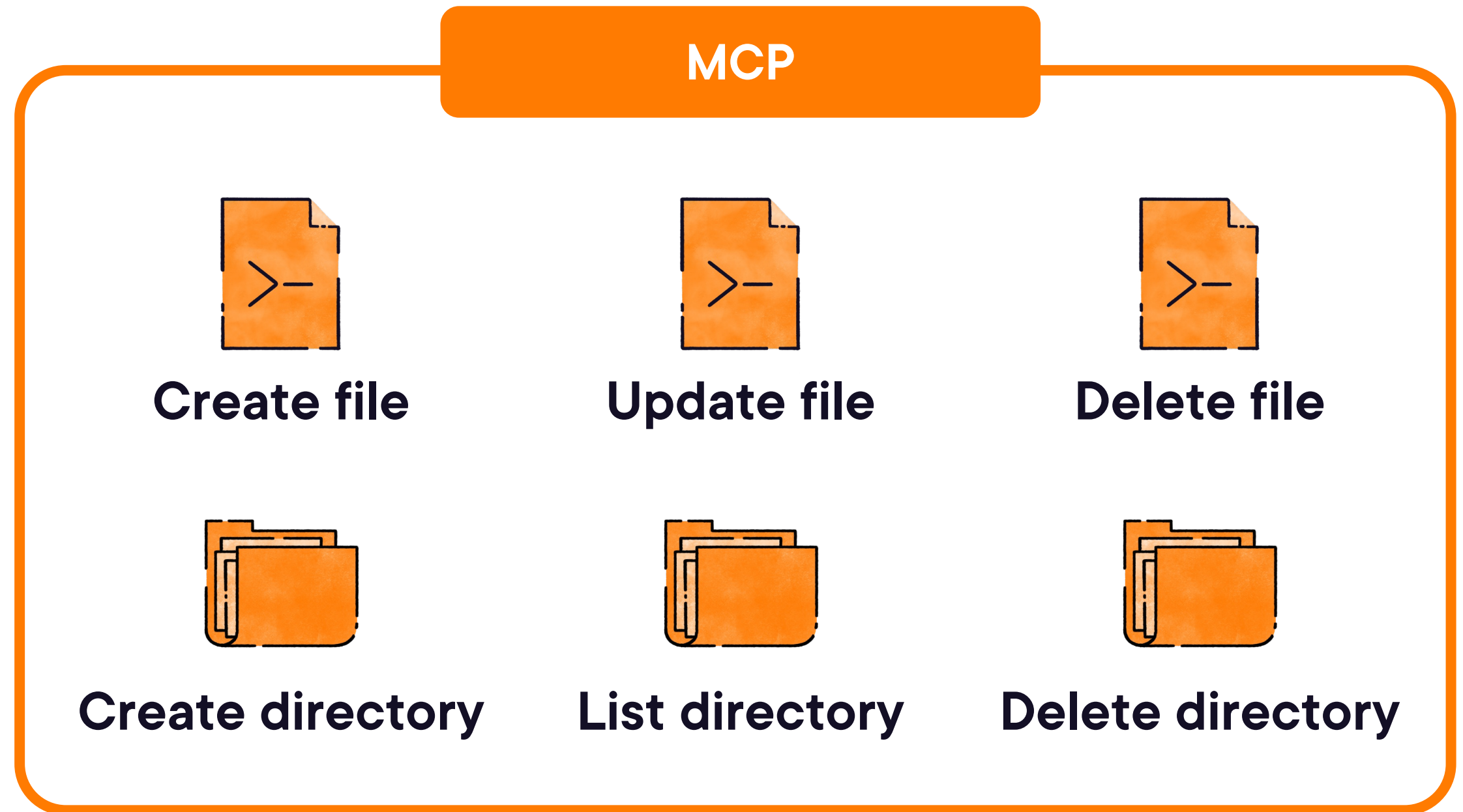
Tools



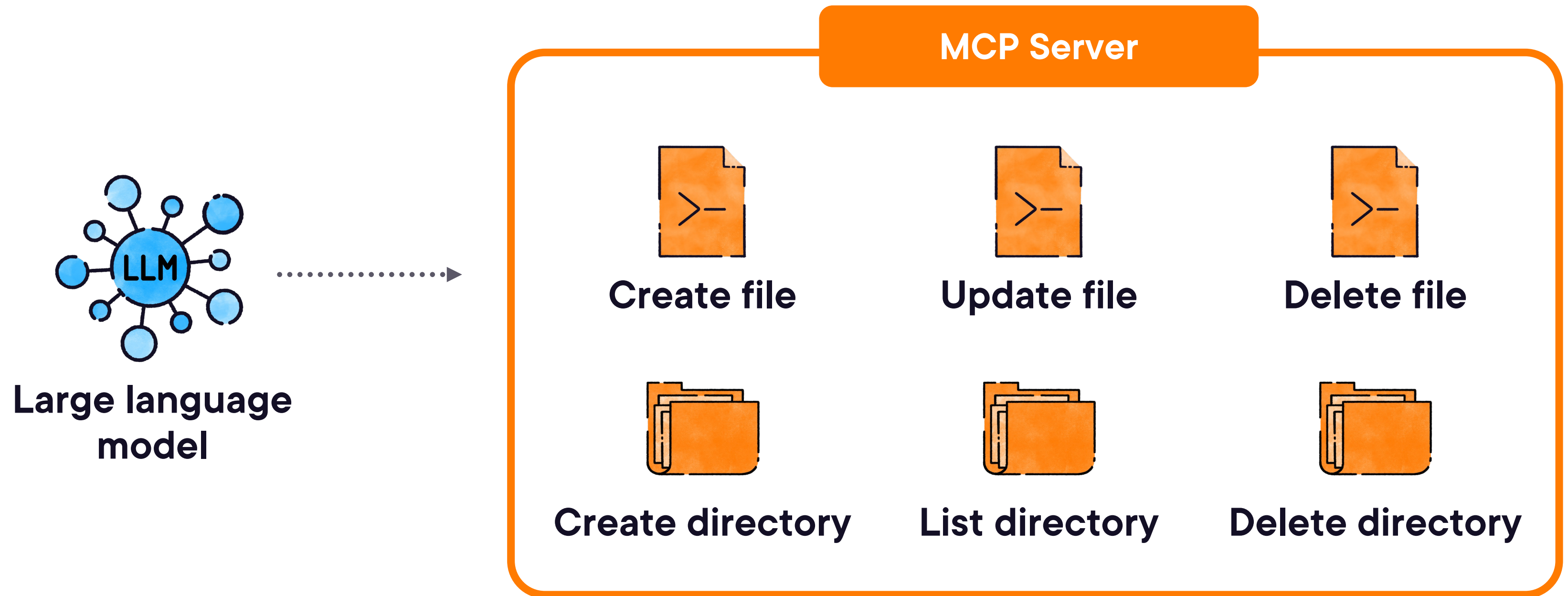
MCP



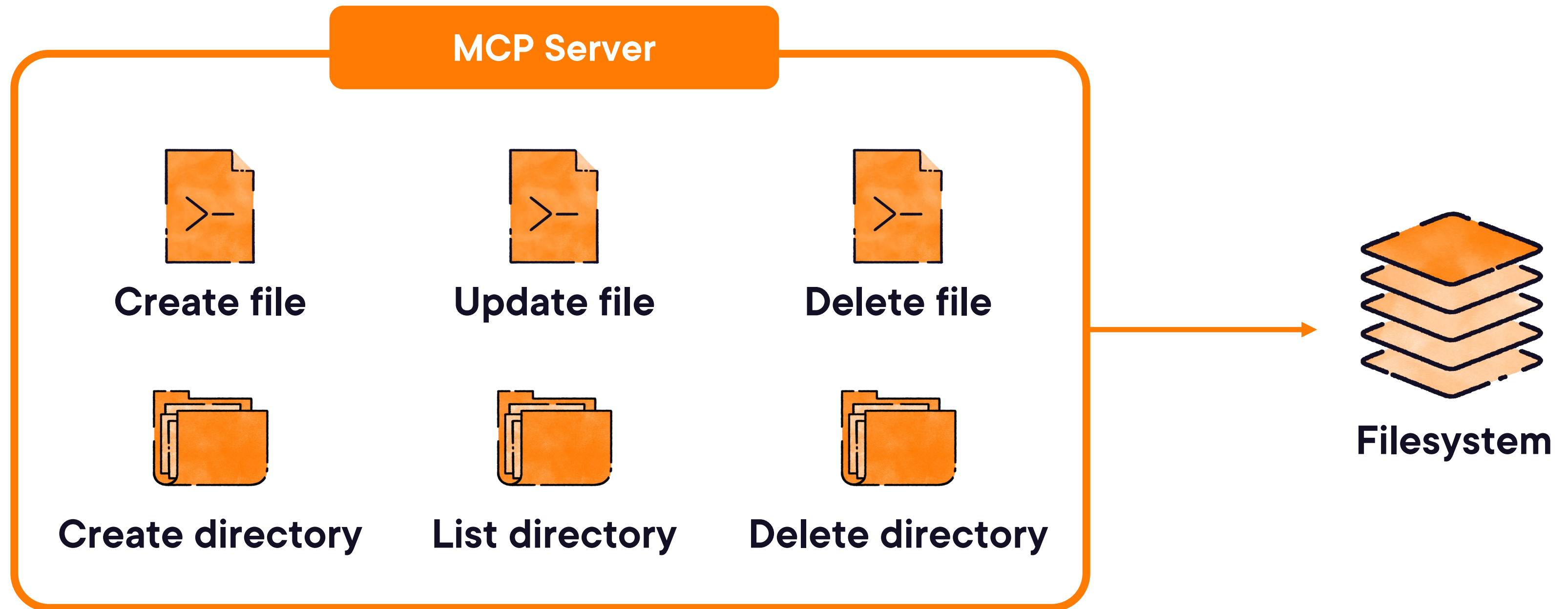
Large language
model



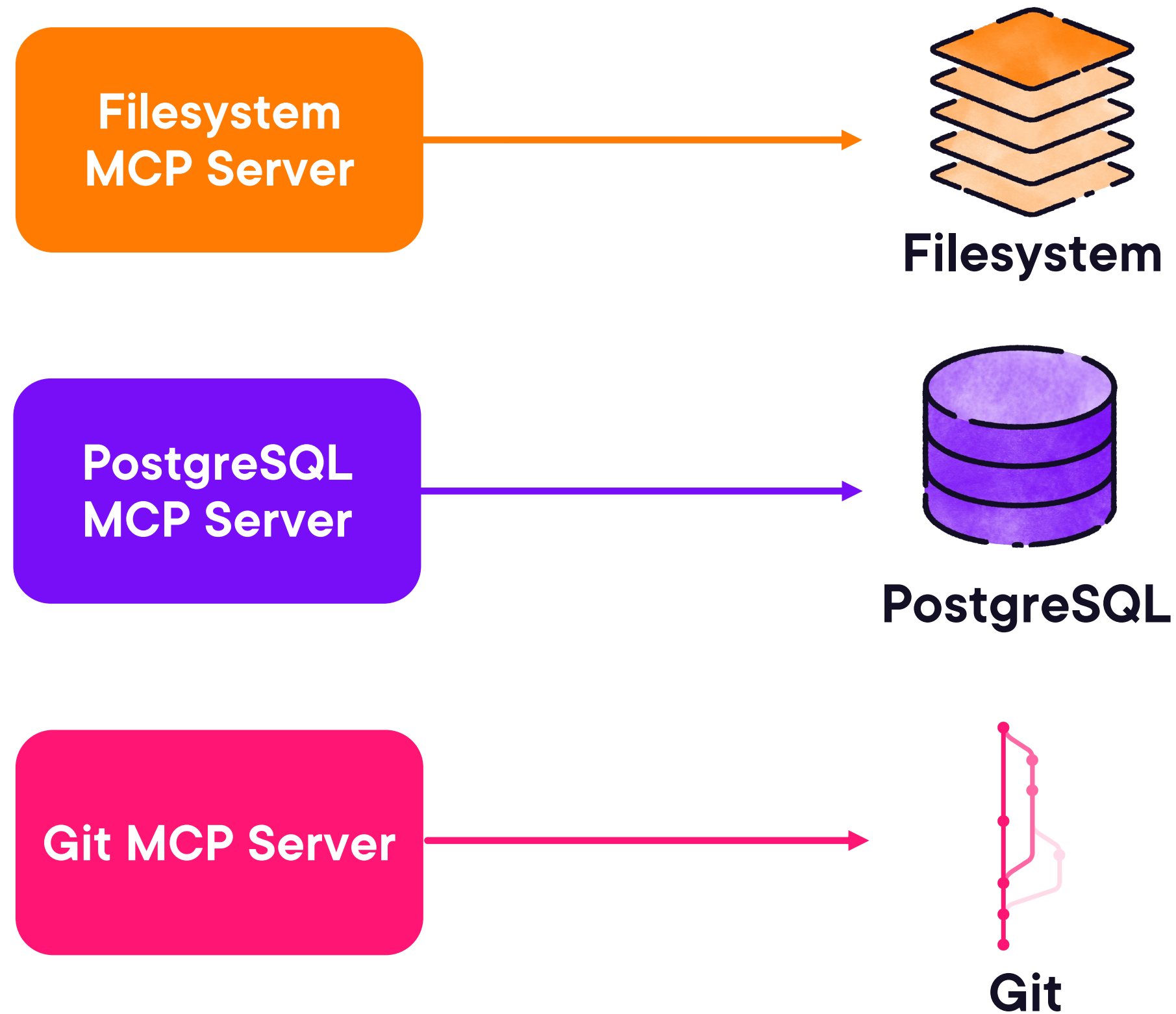
MCP Architecture



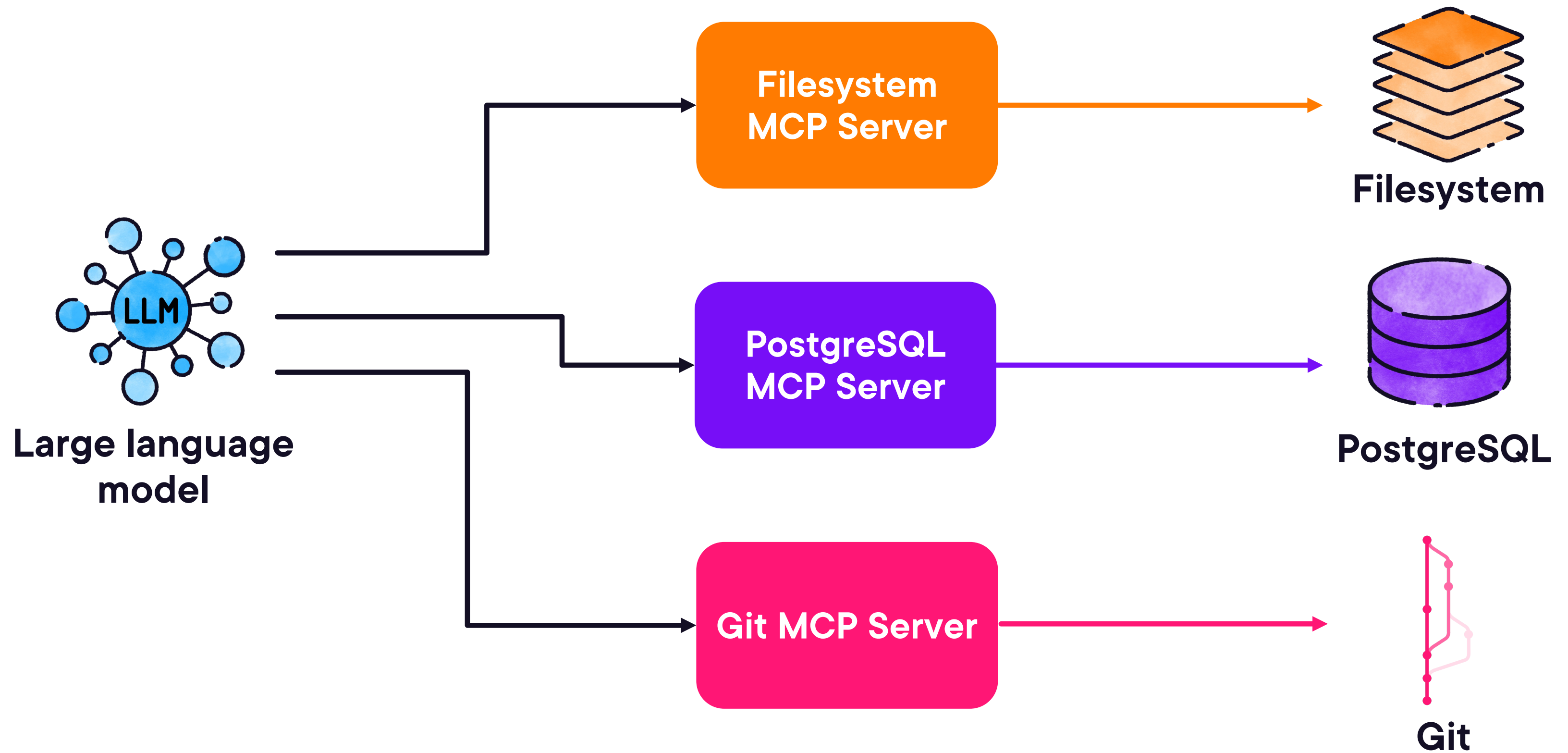
MCP Architecture



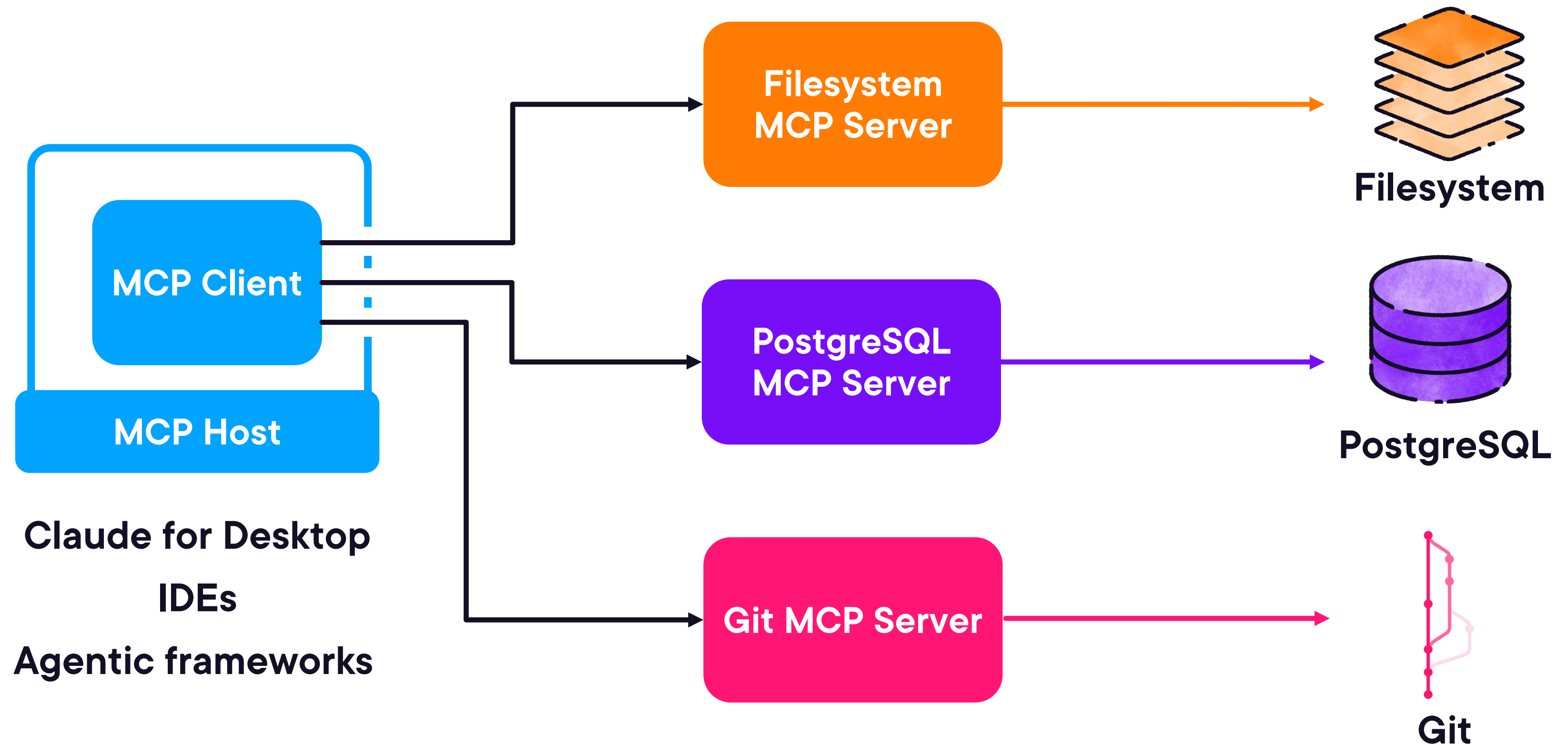
MCP Architecture



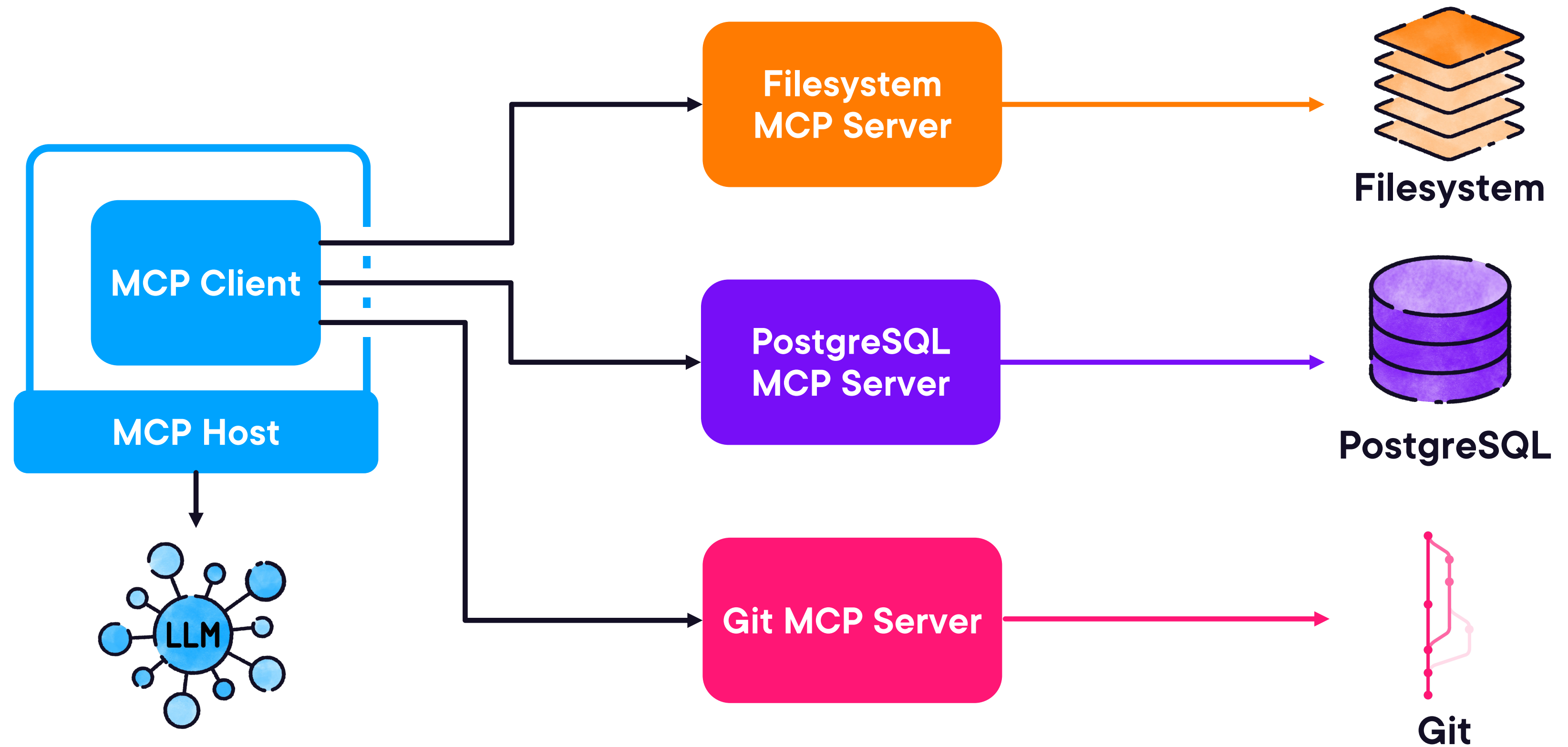
MCP Architecture



MCP Architecture

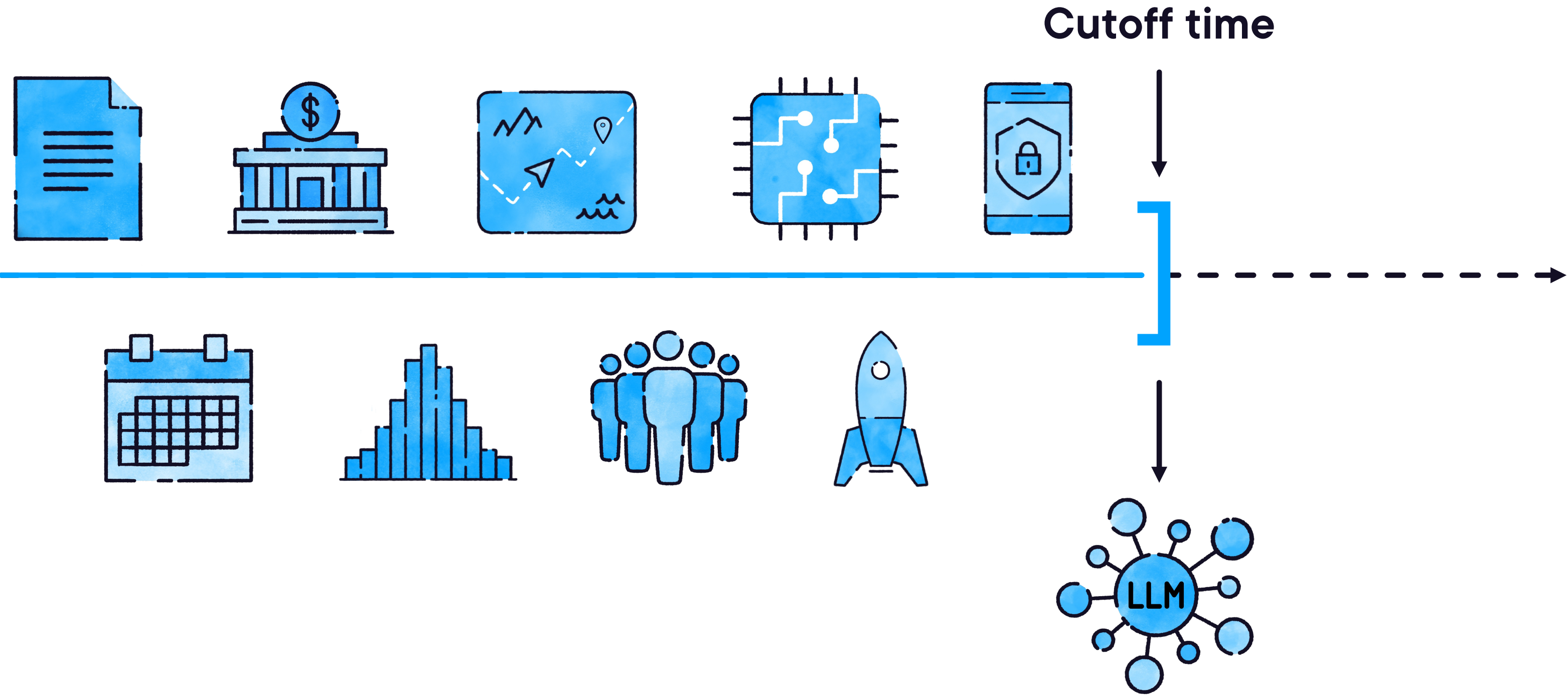


MCP Architecture





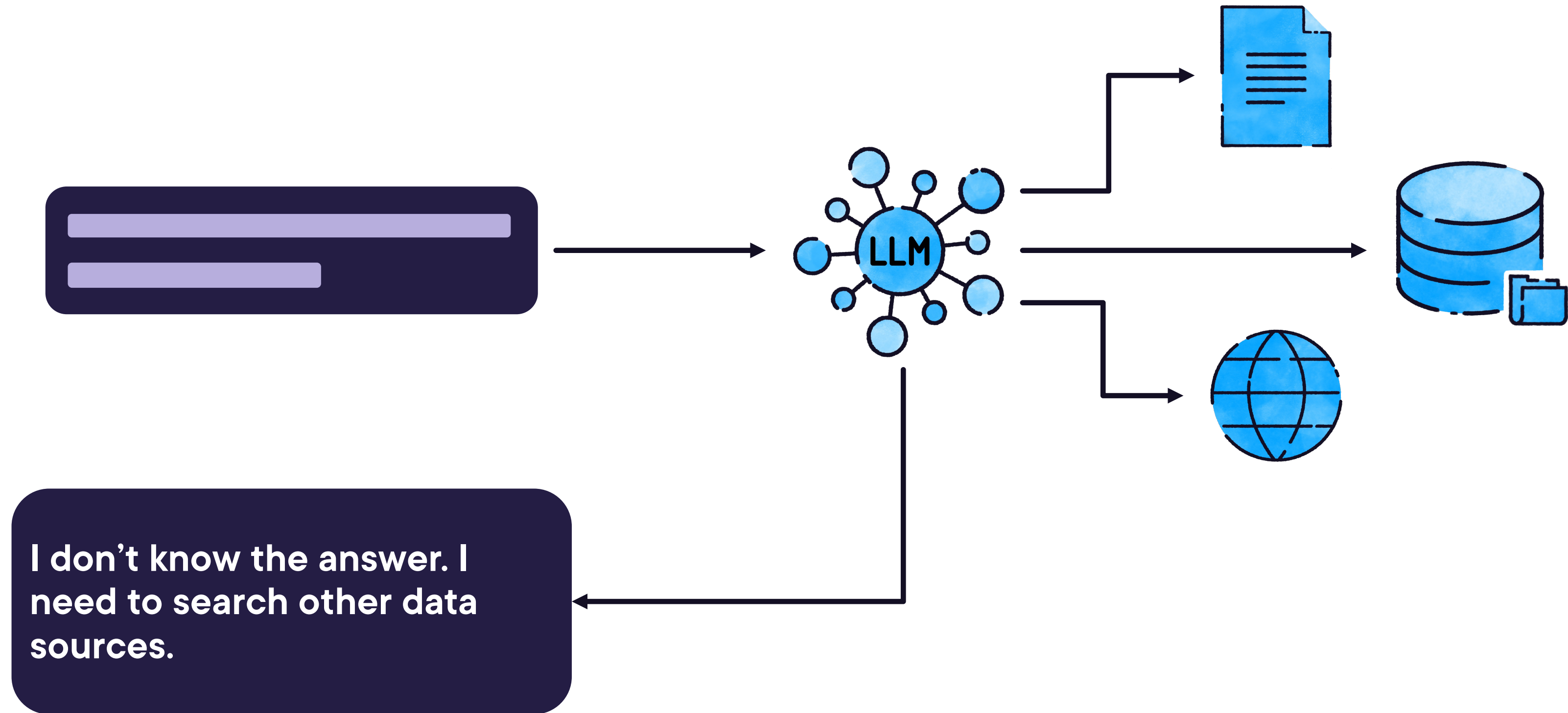
Model Training



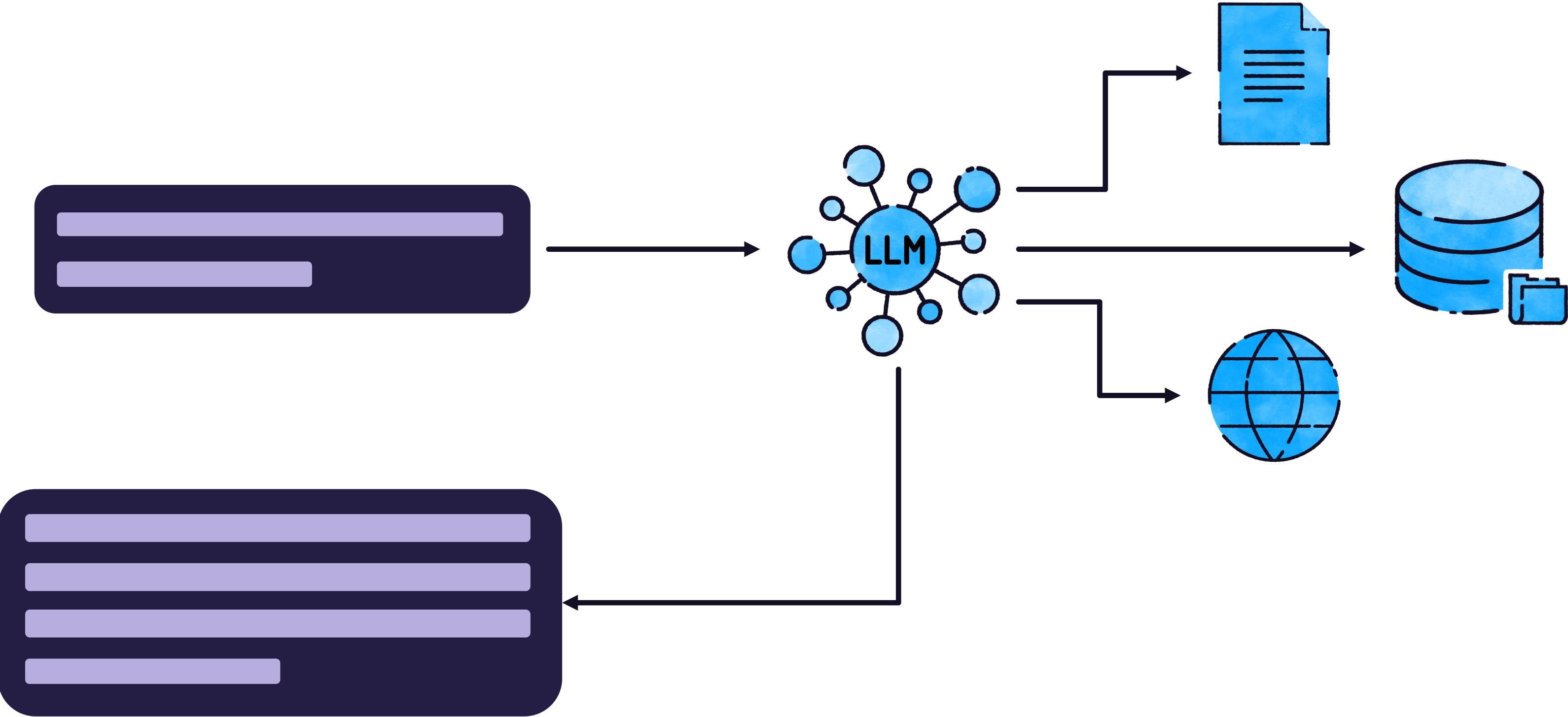
Retrieval Augmented Generation

RAG is a technique that enables large language models (LLMs) to retrieve and incorporate new information.

RAG



RAG



RAG Challenges

Irrelevant retrieval

**Long or complex
documents**

**Security and access
control**



Vector Databases

		Shape	Color	Shininess	Material	Weight	
	[	1.0,	1.0,	1.0,	1.0,	1.0	]
	[	1.0,	0.7,	0.9,	0.4,	0.9	]
	[	1.0,	0.5,	0.7,	0.1,	0.8	]



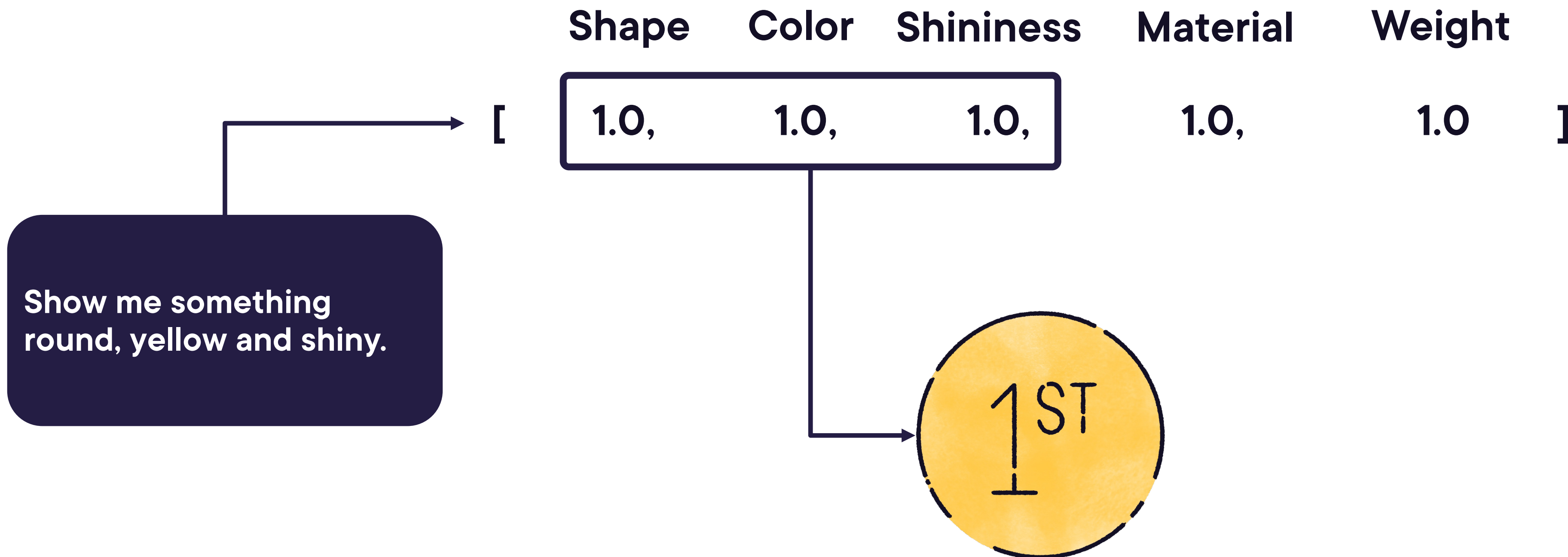
Vector Databases

Show me something
round, yellow and shiny.

	Shape	Color	Shininess	Material	Weight	
[	1.0,	1.0,	1.0,	1.0,	1.0	]
[	1.0,	0.7,	0.9,	0.4,	0.9	]
[	1.0,	0.5,	0.7,	0.1,	0.8	]



Vector Databases





Data Security and Compliance



Risks

Autonomous data access



GitHub MCP Exploit

**Access private
repositories**

Inject malicious code

Leak data



Risks

Autonomous data access

Non-compliant behavior



EU AI Act

Unacceptable risks

Forbidden

High risks

Strict regulation

Limited risks

Transparency regulation

Minimal risks

No regulation

